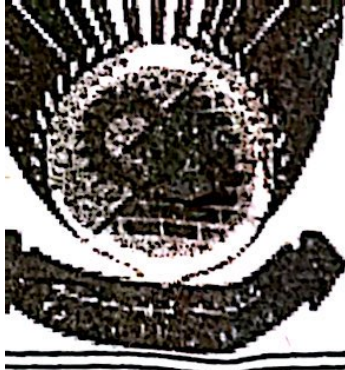




Subject	Logic Design	Date	23.1.2
Level	B2IT/E	Time	2 hours
Max. Marks	50	Signature	
Lecturer	Dr. Mohsen Hussein		

Note: - Question No. **1** is Compulsory.

- Answer any three questions of the rest

Q.1:

a- Draw a circuit diagram for the (unlocked) RS NAND flip-flop and Its truth table.

b- Design combinational logic has 3 inputs (x, y, and z) and 1 output (F). The output (F) is equal 0, when all the inputs are equal to 1, or all input equal 0, or sum of input equal 2.

c- Implement the following function with a multiplexer:

$$F(a, b, c) = \sum(1, 3, 5, 6, 9, 10, 12)$$

Q2: ✓

True or False:

- 1) A NOT gate have more than one input. *False*
- 2) If any input to an OR gate is zero, the output is zero. *False*
- 3) If all inputs to an AND gate are 1, the output is 1. *True*
- 4) A NAND gate can be considered as an AND gate followed by a NOT gate. *True*
- 5) A NOR gate can be considered as an OR gate followed by an inverter. *True*
- 6) The output of an exclusive-OR is 0 if the inputs are same. *True*
- 7) Addition in Boolean algebra is equivalent to the OR function. *True*
- 8) A multiplexer is a logic circuit that allows digital information from a single source to be routed onto several lines. *False*
- 9) A half-adder adds two binary bits *True*
- 10) When the input bits are both 1 and the input carry bit is 1, the sum output of a full adder is 1. ✓
→ True

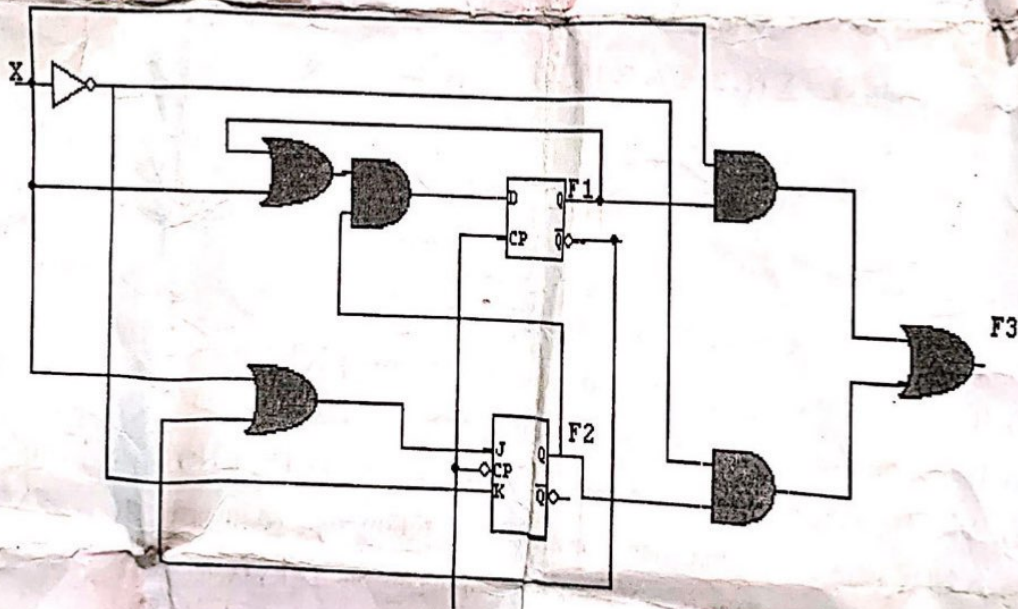
3: ✓

Answers the following:

- When the input to an inverter is HIGH, the output is
(a) HIGH (b) LOW (c) LOW or 1
- The output of an AND gate with inputs A , B and C is 0 (LOW) when
(a) $A = 0, B = 0, C = 0$ (b) $A = 0, B = 1, C = 1$ (c) both answers (a) and (b)
- The output of an OR gate with inputs A , B and C is 0 (LOW) when
(a) $A = 0, B = 0, C = 0$ (b) $A = 0, B = 1, C = 1$ (c) both answers (a) and (b)
- An example of a sum-of-products expression is
(a) $A + B(C + D)$, (b) $AB + AC + ABC$, (c) $(A + B + C)(A + B + C)$
(d) Both answers (a) and (b)
- An example of a product-of-sums expression is
(a) $A(B + C) + AC$, (b) $(A + B)(A + B + C)$, (c) $A + B + BC$, (d) both answers (a) and (b)



Analyze the sequential circuit, (F3=??)



X

How do you get Dec 4*16 by using Dec 3*8?

GOOD LUCK